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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 7: Impacts of Transportation Systems-II

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Contents

- How Transportation impacts Human health ?
- Transport Planning and Mental Health
- Traffic-generated Accidents
- Traffic-generated Air pollution
  - The Case of Los Angeles, California
- Policy Implications
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- Case study
- Conclusion



How Transportation impacts Human health ?

- Transportation has contributed in many ways to improve human life and well-being.

- Benefits includes conveyance of goods and passengers, linking the centers of production, consumption and distribution.



Transportation negatively impacts human health in three broad ways:

- Traffic Accidents
- Environmental pollution
- Spread of Infectious diseases

Source: (Kenneth J. Button, 2010)

How Transportation impacts Human health ?

- In 1989, the International Agency for Research on Cancer concluded that diesel and gasoline engine exhausts are probably carcinogenic to humans.

- In 1998, the California Air Resources Board identified diesel PM as a toxic air contaminant (TAC).

- Children living near heavily travelled roads have about 50% higher risk of suffering from respiratory symptoms from ambient PM_{10} concentrations.



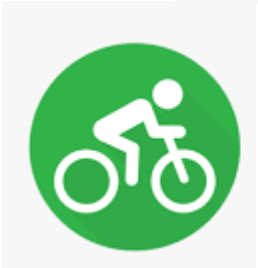
Source: (Kenneth J. Button, 2010)

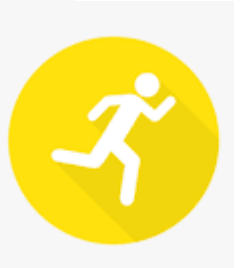
How Transportation impacts Human health ?

- The increased use of private motorcars has contributed to the overall reduction of physical activities.

- Lack of physical activity is one of the major risk factors for coronary heart disease, the leading cause of mortality in Europe.







Walking and Cycling as daily activities can
- promote health
- decrease noise and air pollution

Source: (Kenneth J. Button, 2010)

Transportation-induced Health Impacts





Urban Transport Policy and Planning

Built Environment

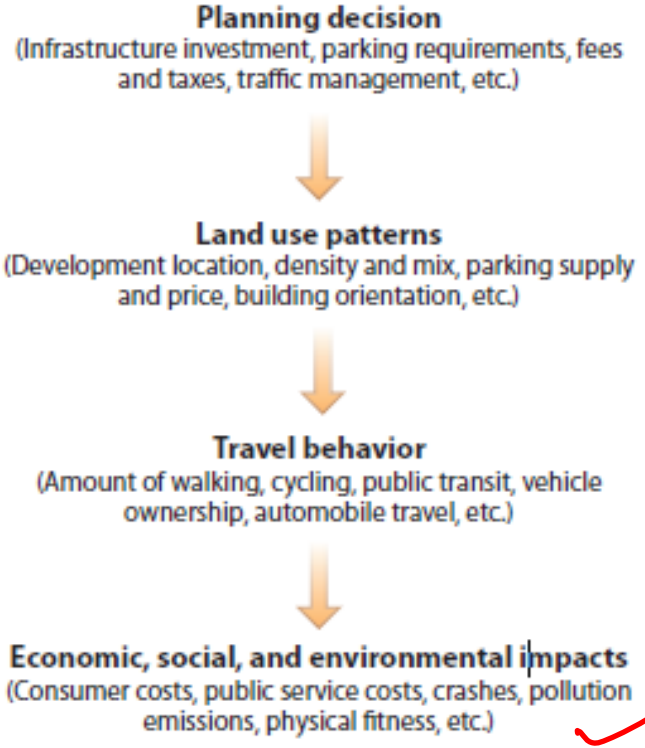
Transport Provision and Mode choice

- Motor vehicle crashes
- Air Pollution
- Noise
- Heat Islands
- Lack of Green space and Biodiversity Loss
- Physical Inactivity
- Social Exclusion
- Community Severance
- Climate Change

- Premature Mortality
- Morbidity

Source: (Haneen Khreis et. al. 2017)

Planning decisions are the main reasons



- Planning decisions can influence land use and travel behaviour, economic, social, and environmental impacts.

Source: (Todd Litman, 2013)

Planning decisions Impacts on Mental Health

- Transport planning decisions can affect mental health
- Improving walking and cycling conditions and public transit tends to improve mental health by
  - increasing physical activity
  - supporting community cohesion (i.e. positive interactions among neighbours)
- Increased neighbourhood walkability
  - reduces symptoms of depression in older men
  - reduces frequency of dementia in women and men
- High-quality public transport facilities can reduce commute stress compared with driving.



Source: (Kenneth J. Button, 2010)

Effects on Mental health

- Children who have the opportunity of playing unhindered by street traffic and without the presence of adults have been found to socially active twice than those who are restricted at homes, unaccompanied by adults because of heavy traffic.



- Lack of this opportunity of children hinders the development of children's independence and reduce their opportunities for social contact and cycling, and is associated with mental ill health, including depression.

Source: (Kenneth J. Button, 2010)

Effects on Mental Health



Effects on mental health, includes

- Risk-taking and aggressive behaviours
- Depression
- Post-traumatic psychological effects of accidents.

- High levels of traffic can cause social isolation and limit interpersonal networks of support, the factors associated with higher mortality and morbidity in the elderly.

Source: (Kenneth J. Button, 2010)

Traffic Accidents

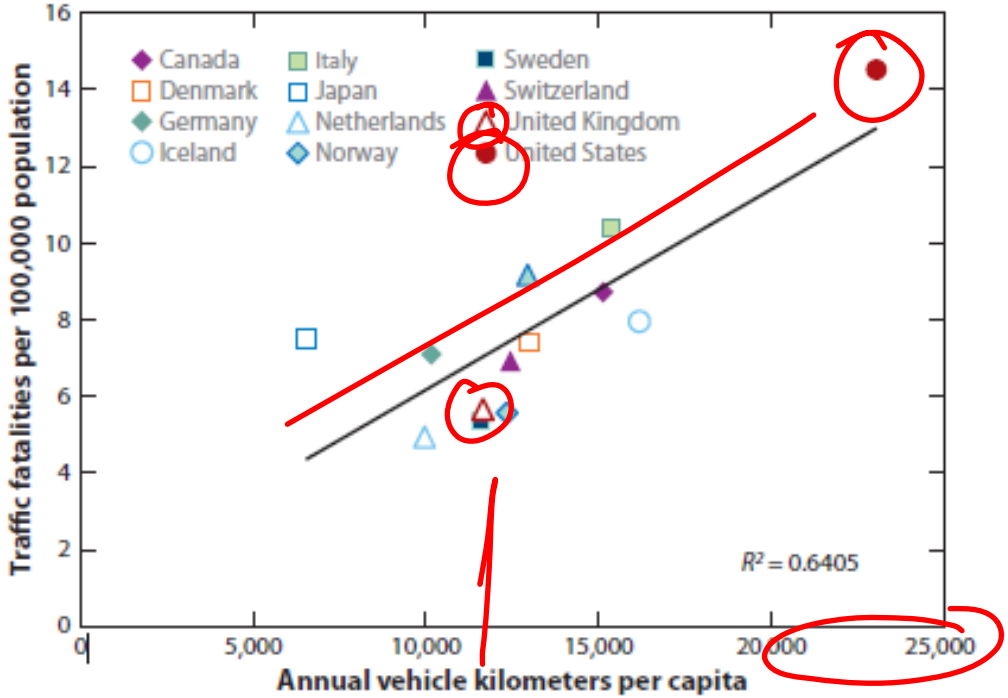
- Globally, every year more than 1.17 million people die in road crashes and over 10 million are crippled or injured.
- Almost 70% of the accidents involves young people.
- India is accounted for about 6% of the global road traffic incidents.
- Accidents resulted in approx. 45,000 deaths in the United States in 2006.





Source: (Kenneth J. Button, 2010)

Vehicle travel and Fatalities



Vehicle travel and traffic fatality rates in OECD countries

Source: (Todd Litman, 2013)

- Among developed countries, per capita traffic fatalities increase with per capita vehicle travel.
- The United States has the highest per capita crash rate.

Role of Traffic-generated Air Pollution

- Motor vehicle traffic is the main source of ground level urban concentrations of air pollutants with recognized hazardous properties.

- CO and benzene levels inside cars can be around 2-5 times higher than at the roadside.

- Car users are more exposed to pollutants than pedestrians, cyclists or public transport users.

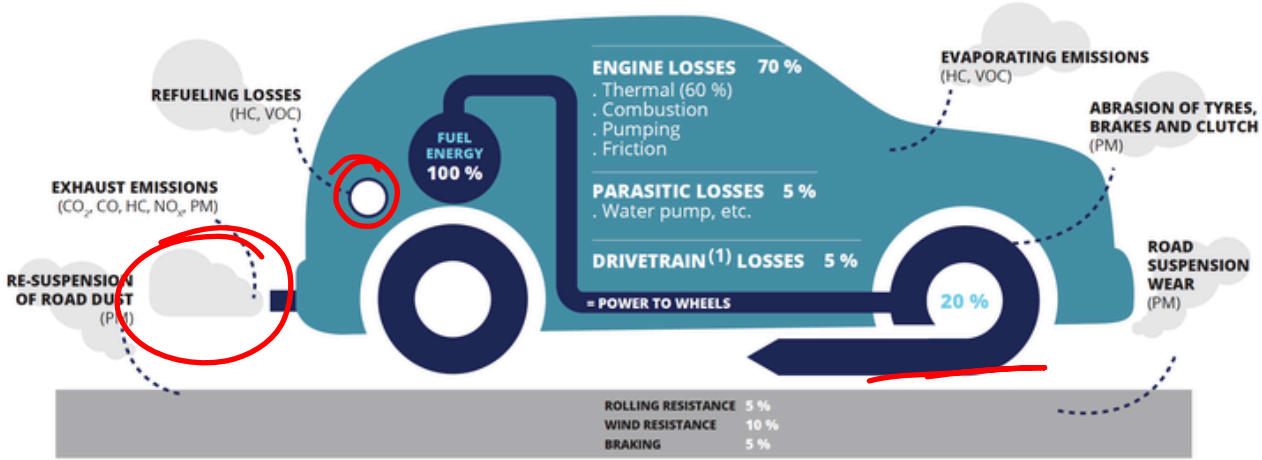


Source: (WHO, 2000)

Role of Traffic-generated Air Pollution

Vehicle emissions and efficiency

Fossil fuel powered road transport represents the most significant source of transport related air pollution. Each vehicle releases pollutants from a number of sources.



(1) The drivetrain of a motor vehicle is the group of components that deliver power to the driving wheels. This includes the transmission, the axles and the wheels.

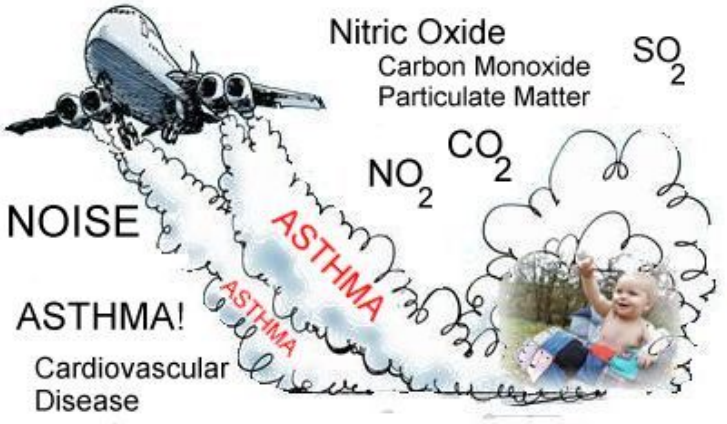
HC - Hydrocarbons; **VOC** - Volatile Organic Compound; **PM** - Particulate Matter; **CO** - Carbon monoxide; **CO_2** - Carbon dioxide; **NO_x** - Nittrogen oxides.

Source: EEA Report — Explaining road transport emissions — a non-technical guide (2016)



Traffic-generated Air pollution

- The main air pollutants from Transport sector includes:
  - Particulate matter (PM_{10} and $PM_{2.5}$)
  - Carbon Monoxide (CO)
  - Nitrogen oxides (NO_x);
  - Non-methane Volatile Organic Carbons (NMVOCs)
  - Ozone (O_3)
  - Benzene
- Trace emissions includes
  - Methane (CH_4),
  - Carbon dioxide (CO_2),
  - Oxides of sulphur (SO_x)
  - Total suspended particles (TSPs)



Source: (WHO, 2006; Ajay Singh Nagpure and Gurjar, B. R, 2012; Bhola Ram Gurjar, 2021)

Traffic-generated Air pollution

Particulate matter (PM_{10})

- Short-term effects of PM_{10} (µm)
  - leads to increased mortality
  - respiratory and cardiovascular diseases
  - increased frequency of respiratory symptoms and use of medication by people with asthma
  - reduced lung function
- Recurrent cumulative exposure increases morbidity and reduces life expectancy.



Source: (WHO, 2000)

The Case of Los Angeles city (LA), California

Exposure to $PM_{2.5}$ pollution



- Millions of California residents live near highways and are exposed to high levels of vehicular air pollution
- People in Los Angeles county are exposed to 60% more vehicular pollution than the state average

Source: (UCUSA, 2019)

The Case of Los Angeles city (LA), California



Source: (https://laist.com/news/happy-smogiversary-la)

Thick Smog in Los Angeles

The thick smog was noticeable over LA from the 1940s and lasted over 3 decades

The Case of Los Angeles city (LA), California



A Smoggy day in Los Angeles, 1964.

LA known as the City of Stars was also called the" City of Smog"

Source: (https://laist.com/news/happy-smogiversary-la)

The Case of Los Angeles city (LA), California



A Smoggy day in Los Angeles, 1966.

People used Gas masks to protect from irritating air

Source: (https://laist.com/news/happy-smogiversary-la)

The Case of Los Angeles city (LA), California

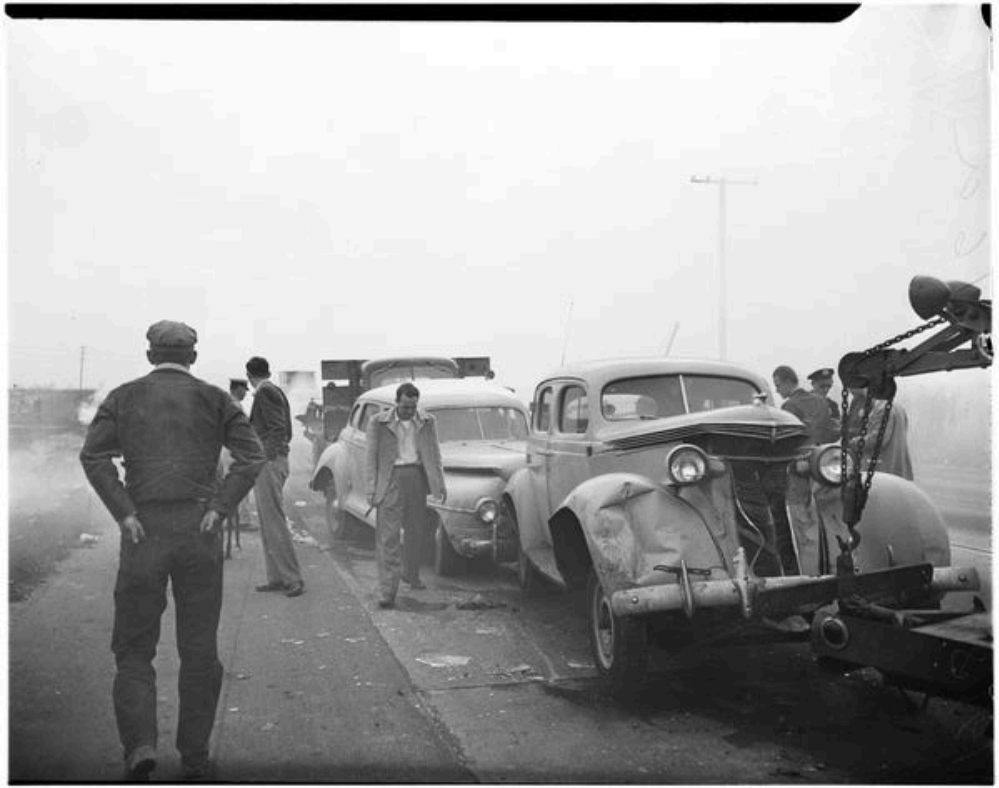


Visibility was very low



Source: (https://laist.com/news/happy-smogiversary-la)

The Case of Los Angeles city (LA), California



Accidents were very common due to low visibility

Source: (https://laist.com/news/happy-smogiversary-la)

The Case of Los Angeles city (LA), California



The Clean Air Act was passed in 1970

Harsher pollution control measures were implemented in LA and California

Source: (https://laist.com/news/happy-smogiversary-la)

The Case of Los Angeles city (LA), California



Smog in Los Angeles, 1972



Source: (https://laist.com/news/happy-smogiversary-la)

A Smoggy day in Los Angeles



Los Angeles used to be the most polluted air basin in the United States.

Source: (California Air Resources Board)

A Clear day in Los Angeles



Source: (California Air Resources Board)

Determinants of Traffic-generated Air pollutants

- The levels and composition of pollutants in the air depends on
  - Number of vehicles
  - Age of vehicles
  - Engine type and condition
  - Type of fuels used
  - Meteorology
  - Shape of the urban environment
  - Traffic organization



Source: USDOT Transportation and Climate Change Clearinghouse Modal Wheel. Developed by ICF International

Source: (WHO, 2000)

Determinants of Traffic-generated Air pollutants

- Heavy vehicles are major polluters.
  - Ex. When compared to a car with a catalytic converter, a diesel truck produces 50-100 times more fine and ultra-fine particles per km travelled.
- Modern diesel engines emits less $PM_{2.5}$ but a larger number of ultra-fine particles than older engines.
- It is now suspected that the ultra-fine particles (or the number of particles, rather than particle mass) are the main cause of health effects in humans.
- If this is true, then the new diesels cause more harm than the old ones.

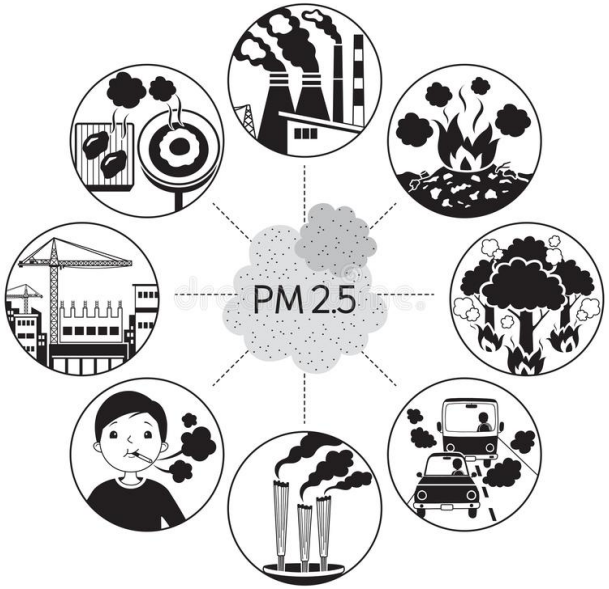


Source: (WHO, 2000)

Health Effects of Air Pollution

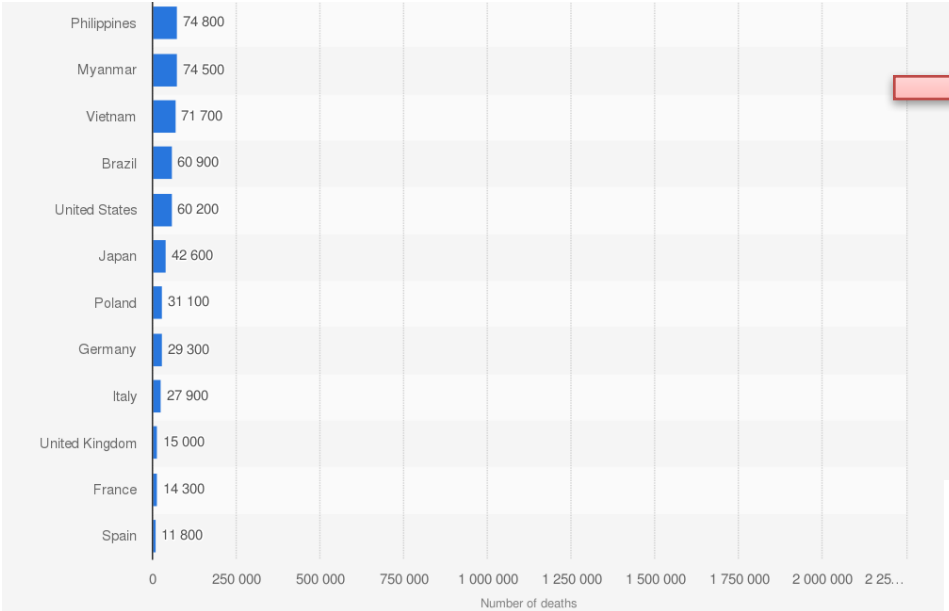
Health end-points for PM and other exposure can cause

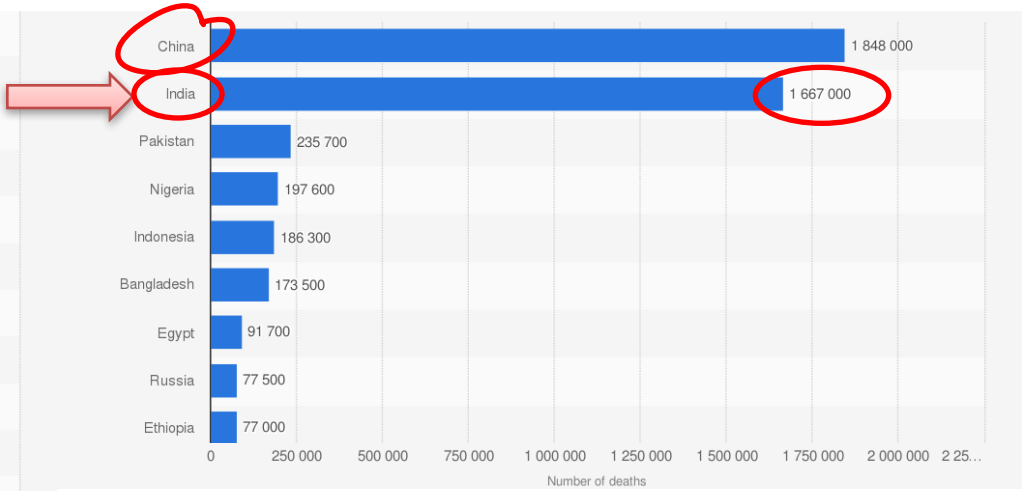
- Mortality (aged ≥30 years, excluding accidental causes) – long term
- Cardiovascular diseases
- Respiratory diseases
- Acute bronchitis (aged <15 years)
- Asthma exacerbation (aged <15 years)
- Asthma exacerbation (aged ≥15 years)
- Restricted physical activity (aged 15–64 years)
- Lung cancer



Source: (WHO, 2006)

Fatalities due to Air Pollution

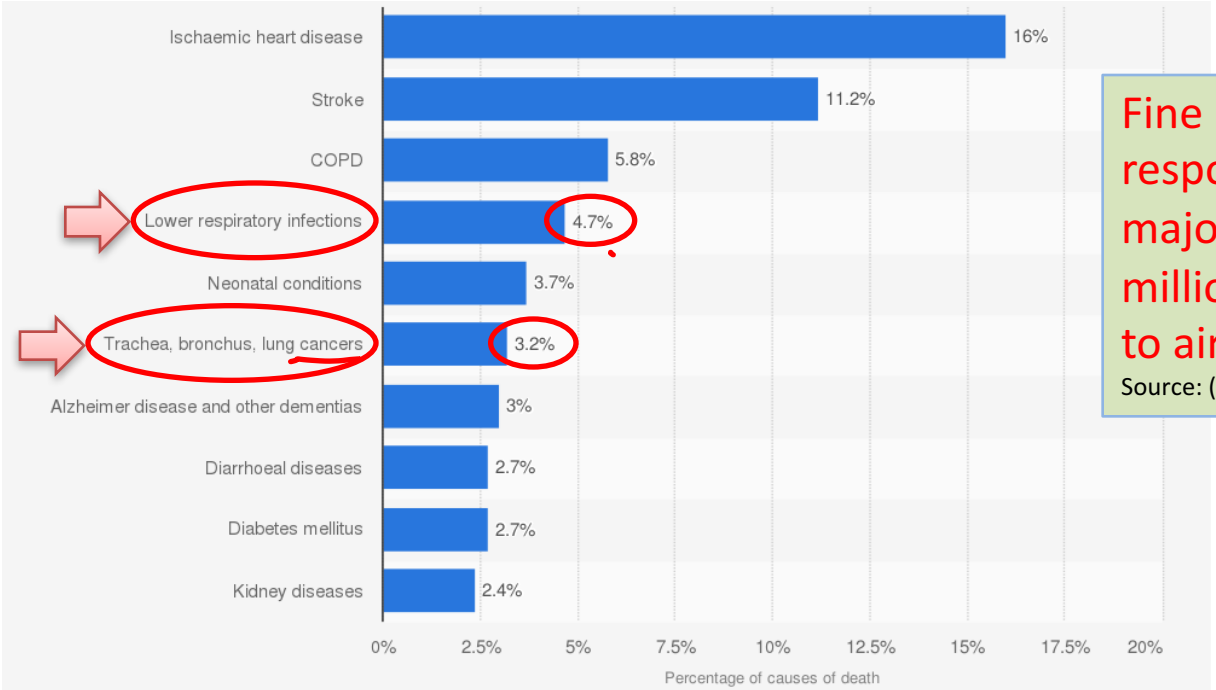




Number of Deaths by Country due to Air Pollution Worldwide, 2019

Source: (IHME, Health Effects Institute, 2020)

Leading causes of Death Worldwide



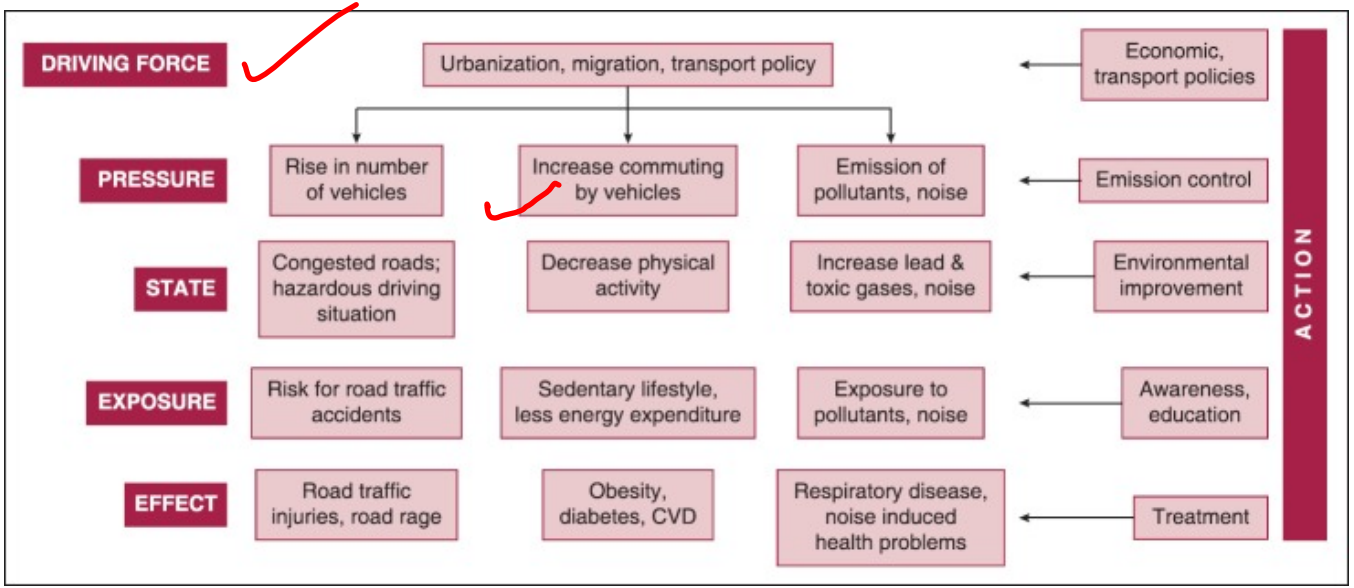
Fine particulate air pollution is responsible for the vast majority of the 3 million to 4 million annual deaths attributed to air pollution worldwide.

Source: (UCUSA, 2019)

Percentage of leading 10 causes of death worldwide in 2019

Source: (WHO, 2021)

Mitigating the effects of Transport Sector on Human health



The DPSEEA Framework, WHO

- The Driving force, Pressure, State, Exposure, Effect and Action (DPSEEA) framework of the World Health Organization

Source: (Hariom Kumar Solanki et., al. 2016)

Health Improvement Strategies

Policy Implications



- Technological improvements, such as the introduction of unleaded petrol and catalytic converters, have already had a positive impact
- A holistic approach should be adopted.
- Tackling individual pollutants in isolation could even be counterproductive if it leads to increases in another pollutant component.
  - Ex. Attempting to lower CO_2 emissions through the promotion of new diesel vehicles, would lead to an increased number of ultra-fine particles, which seem themselves to be a cause of concern

Source: (WHO, 2000)

Health Improvement Strategies

Co-Benefits Policy Implications

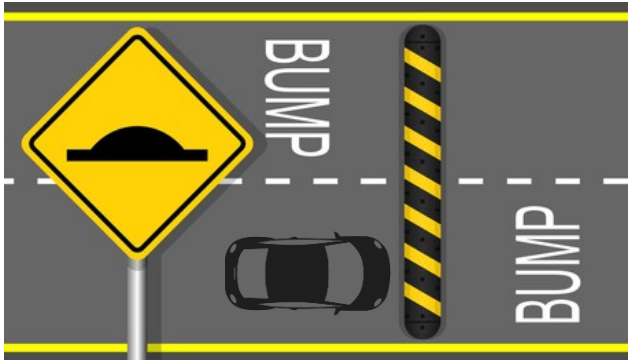
- The effect on the pollution mix as a whole must always be considered in designing interventions.
  - Ex. In selecting actions to reduce emissions of greenhouse gases, those that also reduce other air pollutants, such as particulate matter, should have priority.



Source: (WHO, 2000)

Health Improvement Strategies

Traffic Calming and Speed Control



- Improves safety by
  - Reducing crash frequency and severity
  - Reduce total vehicle travelled

- Impacts on pollution emissions vary on conditions
  - Ex. Per-km emission rates tend to decline with reductions from high to moderate traffic speeds, but rates increase at very low speeds (under 20 km per hour) and with more stop and-go driving (for example, due to speed bumps).

- Speed reductions tend to improve walking and cycling conditions, which can reduce per capita emissions, increase physical activity, and improve basic access.

Source: (Todd Litman, 2013)

Health Improvement Strategies



Active Transport Improvements (Walking and Cycling)

- Walking and cycling infrastructure improvements, such as better sidewalks, crosswalks, and paths, can reduce crash risks.

Increasing active transport can
- Reduce total crash rates
- Large reductions in air pollution emissions
- Reduce congestion



- A 1% shift from motorized to Non-motorized Transport (NMT) modes can typically reduce emissions by 2–4%.
- Also increases physical activity

Source: (Todd Litman, 2013)

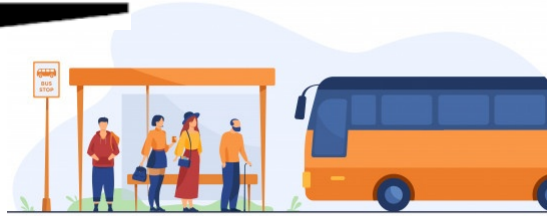
Health Improvement Strategies

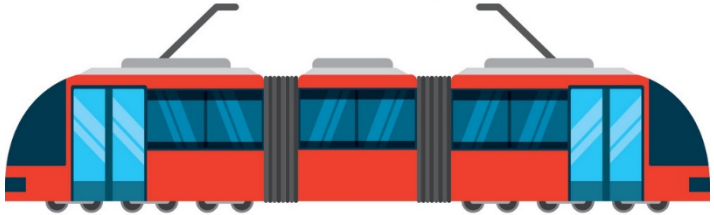
Public Transit Service Improvements

- Public transport service improvements can reduce Total crash rates and pollution emissions.

- Public transport service can be improved by
  - Providing more routes
  - Longer operating hours
  - Frequent transit services
  - Quality vehicles and terminal stations
  - Grade separations
  - Real-time user information systems







Public transit travel has lower traffic fatality rate than other automobile travel

Source: (Todd Litman, 2013)

Health Improvement Strategies

Transport Pricing Reforms

- A one-cent state gasoline tax increase in the U.S reduces per capita traffic fatalities 0.25% and traffic fatalities per vehicle-mile by 0.26% (Grabowski & Morrisey, 2006).

- A sustained 20% gasoline price increase in the U.S would prevent ≈2,000 traffic crash deaths (≈5% of the total), plus ≈600 air pollution deaths (Leigh & Geraghty, 2008).

- A 1% vehicle travel reduction reduces crashes by more than 1% (Chi et al., 2010).

Pricing reforms increases use of active modes and therefore physical fitness.



Source: (Todd Litman, 2013)

Health Improvement Strategies

Mobility Management Marketing

- Mobility management marketing refers to various programs and information resources that encourage people to shift travel from automobiles to alternative modes.

- Methods include:
  - Commute trip reduction programs (employers encourage their employees to use alternative modes)
  - Transportation management associations (through which businesses support alternative modes)
  - Ride matching and Vanpool support programs
  - Direct marketing programs



Source: (Todd Litman, 2013)

Health Improvement Strategies

Smart Growth Land Use Development Policies

- Smart growth policies encourage
  - More compact and mixed development
  - More connected path and road networks
  - Better integration between transport and land use planning
  - Improved walkability
  - More efficient parking management
  - Other features that improve accessibility and transport diversity



Source: (Todd Litman, 2013)

Health Improvement Strategies

Smart Growth Land Use Development Policies

- People who live and work in Smart growth communities tend to own fewer motor vehicles, drive less, and rely more on walking, cycling, and public transport.

- Smart growth residents typically drive 20–40% less than automobile-dependant communities.

- Smart growth residents have substantially lower per capita traffic casualty rates than automobile-dependant communities.



Source: (Todd Litman, 2013)

Health Improvement Strategies

Smart Growth Land Use Development Policies



- Smart growth communities reduce pollution emissions, although more compact development may increase some pollution exposure
  - Ex. If more people walk, bike, live, and work close to busy roadways create a canyon effect on urban roads.

- These risks can be mitigated through targeted strategies such as using cleaner transit vehicles on major urban roads and locating sidewalks and paths away from traffic.

Source: (Todd Litman, 2013)

Health Improvement Strategies

Smart Growth Land Use Development Policies

- Smart growth community increases active transport significantly because more destinations are within walking and cycling distances

Smart Growth communities include improvements to walking, cycling, and public transport such as better sidewalks and crosswalks, traffic calming, bike and bus lanes, and bike racks.



Source: (Todd Litman, 2013)

Case Study: Potential Health Risks due to in-car aerosol exposure across 10 Global cities

$PM_{2.5}$ Data Collection

- $PM_{2.5}$ data were collected every 1 minute, using a portable laser particle counter, from the backseat of a passenger car.

- 3 car settings were used:
  - Windows-open
  - Windows-closed with fan on
  - Windows-closed with recirculation

- Morning peak, Evening peak and Off-peak data was collected.



Portable particle counter

Source: (Prashant Kumar et. al, 2021)

Case Study: Potential Health Risks due to in-car aerosol exposure across 10 Global cities

Hotspot Identification

- Hotspots in 10 cities were identified by counting data points exceeding 90th percentiles (P90) of the one-minute average $PM_{2.5}$ concentrations during window-open setting.

- $PM_{2.5}$ concentrations were used to compare the inhaled doses at hotspots to those concentrations at the free-flow segments of roads.

- $PM_{2.5}$ concentrations were also compared to socio-economic parameters such as GDP, value of statistical life and road density, to derive the health burden and economic losses.

Source: (Prashant Kumar et. al, 2021)

Case Study: Inference

- A relatively short time spent at hotspots contributed to a significant concentration of total inhaled air pollution doses, during a car travel.
  - Ex. Hotspots at cities Guangzhou (CAN) and Addis Ababa (ADD) accounted for only about 26-28% of the total time spent on each route, but corresponded to around 54-56% of the total inhaled doses during a trip.
- An increase in percentage length of hotspots were observed with the increase in road density.
- The percentage length of hotspots shows negative correlation with increasing GDP and Value of Statistical life (VSL), except for the city Guangzhou (CAN).

Source: (Prashant Kumar et. al, 2021)

Conclusion

- The Negative impacts of Transportation on human health can vary from Respiratory and Cardiovascular diseases to Lung Cancer to Death.

- Health Improvement Strategies should be given primary concern from the planning stage of Transportation systems.

  - Focus on strategies such as
    - Strict and effective Policy implications
    - Promoting Active transport
    - Improvements in existing public transit facilities
    - Pricing reforms
    - Mobility management awareness measures
    - Smart growth land use development measures.

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Thank You